# ASSIGNMENT-3.5

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**Question 1**: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks

whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.

**CODE and Output:**

```python
# Q1: Leap Year Check
year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"{year} is a leap year")
else:
    print(f"{year} is not a leap year")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downl
env/4.py"
Enter a year: 2024
2024 is a leap year
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downl
env/4.py"
Enter a year: 2026
2026 is not a leap year
```

Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python

function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

• Compare with a zero-shot solution.

• Analyze algorithm efficiency.

**CODE and Output :**

```python
# Q2: GCD of Two Numbers
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

num1 = 12
num2 = 18
print(f"The GCD of {num1} and {num2} is {gcd(num1, num2)}")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downl
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHAR
env/4.py"
Enter a year: 2024
2024 is a leap year
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHAR
env/4.py"
Enter a year: 2026
2026 is not a leap year
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHAR
env/4.py"
The GCD of 12 and 18 is 6
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHAR
env/4.py"
```

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases.

```python
# Q3: LCM Calculation
def lcm(a, b):
    def gcd(x, y):
        while y:
            x, y = y, x % y
        return x
    return abs(a * b) // gcd(a, b)

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))
print(f"The LCM of {num1} and {num2} is {lcm(num1, num2)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI
env/llab.py"
Enter first number: 4
Enter second number: 6
The LCM of 4 and 6 is 12
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI
env/llab.py"
Enter first number: 3
Enter second number: 9
The LCM of 3 and 9 is 9
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

- Test with valid and invalid binary inputs.
- Identify missing validation logic.

```python
# Q4: Binary to Decimal Conversion
binary_num = input("Enter a binary number: ")
decimal_num = int(binary_num, 2)
print(f"The decimal equivalent of binary {binary_num} is {decimal_num}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI
env/llab.py"
Enter a binary number: 1010
The decimal equivalent of binary 1010 is 10
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI
env/llab.py"
Enter a binary number: 1110
The decimal equivalent of binary 1110 is 14
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function

that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

```python
# Q5: Decimal to Binary Conversion
decimal_num = int(input("Enter a decimal number: "))
binary_num = bin(decimal_num).replace("0b", "")
print(f"The binary equivalent of decimal {decimal_num} is {binary_num}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.
env/llab.py"
Enter a decimal number: 101
The binary equivalent of decimal 101 is 1100101
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.
env/llab.py"
Enter a decimal number: 1101
The binary equivalent of decimal 1101 is 10001001101
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks

whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

```python
# Q6: Harshad Number Check
num = int(input("Enter a number: "))
sum_of_digits = sum(int(d) for d in str(num))
if num % sum_of_digits == 0:
    print(f"{num} is a Harshad Number")
else:
    print(f"{num} is Not a Harshad Number")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Down
env/llab.py"
Enter a number: 18
18 is a Harshad Number
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Down
env/llab.py"
Enter a number: 46
46 is Not a Harshad Number
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```